# Lab Mid Exam Solution: HOG vs SIFT on Blood-Cell Images

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## Personal Settings & Fixed Configuration

| ***Setting*** | ***Rule*** | ***Your Value*** |
|---|---|---|
| Random seed | Last two digits of your registration number | 17 |
| HOG cell size | Last digit even: 8 x 8. Last digit odd: 16 x 16 | 16 x 16 (last digit is 7, which is odd) |
| SIFT grid size | Last digit 0 to 4: 4 x 4 grid (16 keypoints). Last digit 5 to 9: 6 x 6 grid (36 keypoints) | 6 x 6 = 36 keypoints (last digit is 7) |

- Dataset: BloodMNIST (size=64), resized to 128 x 128 grayscale (uint8)
- Kept Classes: eosinophil (1), monocyte (5), neutrophil (6), platelet (7), mapped to class IDs 0, 1, 2, 3
- Subset Sizes: 100 train, 30 validation, and 30 test images per class

## *Q1. Data Setup (3 Marks)*

### 1. Full Dataset Split Shapes

- Train full split: (11959, 64, 64, 3), labels: (11959, 1)
- Validation full split: (1712, 64, 64, 3), labels: (1712, 1)
- Test full split: (3421, 64, 64, 3), labels: (3421, 1)

### 2. Filtered & Preprocessed Subset Shapes and Class Counts

- Train Split: X shape = (400, 128, 128), dtype=uint8, y shape = (400,)
  Class counts: {'eosinophil': 100, 'monocyte': 100, 'neutrophil': 100, 'platelet': 100}
- Validation Split: X shape = (120, 128, 128), dtype=uint8, y shape = (120,)
  Class counts: {'eosinophil': 30, 'monocyte': 30, 'neutrophil': 30, 'platelet': 30}
- Test Split: X shape = (120, 128, 128), dtype=uint8, y shape = (120,)
  Class counts: {'eosinophil': 30, 'monocyte': 30, 'neutrophil': 30, 'platelet': 30}

### 3. Example Image Grid

A 4x4 grid showing 4 training examples per class in 128x128 grayscale:

## Q2. HOG + SVM (6 Marks)

### (2) HOG Feature Length Calculation by Hand

Image size: 128 x 128 pixels
Cell size: 16 x 16 pixels (registration number ends in 7 -> odd)
Cells per image: 128/16 = 8 -> 8x8 cells
Block size: 2 x 2 cells per block
Blocks per side: 8 - 2 + 1 = 7
Total blocks: 7 x 7 = 49 blocks
Features per block: 4 cells/block * 9 orientations = 36 values per block
Total HOG Feature Vector Length: 49 blocks * 36 values/block = 1764 features

### (1) HOG Visualization

### (3) SVM Training and Validation Results

Best C chosen on validation set: 0.1
Validation Accuracy: 0.7583
Validation Macro-F1 Score: 0.7561

Confusion Matrix:
[[19 8 3 0]
[ 4 22 3 1]
[ 6 3 20 1]
[ 0 0 0 30]]

## Q3. SIFT + SVM (7 Marks)

### (2) Normal SIFT Detection

Average keypoints per image: 72.00
Minimum keypoints per image: 11
Maximum keypoints per image: 184
Images with zero keypoints: 0 of 400

Why a different number of keypoints per image is a problem for an SVM: An SVM classifier needs every input image to be represented by a fixed-length feature vector of the exact same size, but normal SIFT finds a different number of keypoints in each image, which creates feature vectors of unequal lengths that cannot be passed directly into an SVM.

### (2) Fixed Grid SIFT Feature Length Calculation by Hand

Grid size: 6 x 6 = 36 keypoints (last digit is 7, between 5 and 9)
Descriptor size per keypoint: 128 values
Total SIFT Feature Vector Length: 36 grid points * 128 = 4608 features

### (3) SVM Training and Validation Results

Best C chosen on validation set: 0.1
Validation Accuracy: 0.7083
Validation Macro-F1 Score: 0.7013

Confusion Matrix:
[[16 9 4 1]
[ 4 24 1 1]
[ 8 7 15 0]
[ 0 0 0 30]]

## Q4. Robustness (6 Marks)

### (1) Prediction Before Running Test Evaluation

Which model will lose more accuracy under rotation (20 degrees): I predict HOG will lose more accuracy under rotation because HOG's 20-degree-wide orientation bins and fixed cell layout shift when the image turns.

Which model will lose more accuracy under noise: I predict HOG will lose more accuracy under noise because HOG takes gradients directly from raw pixels, so noise adds random gradient energy in every cell.

### (4) Test Set Evaluation Table

| *Model* | *Clean accuracy* | *Rotated accuracy (20°)* | *Noisy accuracy (sigma=0.05)* |
|---|---|---|---|
| HOG + SVM | 0.7333 | 0.6917 | 0.2500 |
| SIFT + SVM | 0.7167 | 0.7500 | 0.7083 |

### (1) Was Your Prediction Correct?

Under rotation HOG fell by 4.2 points and SIFT by -3.3 points, so HOG lost more. Under noise HOG fell by 48.3 points and SIFT by 0.8 points, so HOG lost more. My rotation prediction (HOG) was correct and my noise prediction (HOG) was correct. A likely reason is that HOG's 20-degree-wide orientation bins and fixed cell layout shift when the image turns, and for noise that HOG takes gradients directly from raw pixels, so noise adds random gradient energy in every cell.

## Q5. Short Answers and AI Log (3 Marks)

### (1) Short Conceptual Questions

***(a) Why does HOG use block normalisation?***

HOG uses block normalisation to group nearby cells together (2x2 cells per block) and divide their gradient values by the overall magnitude of the block. This makes the features much less sensitive to changes in lighting, shadows, or image contrast.

***(b) Why is it wrong to choose C using the test set?***

If we use the test set to pick the best C, we are leaking test information into our model selection process and overfitting the test data. The validation set is meant for tuning hyperparameters like C, while the test set should only be used once at the end to check final performance.

## (2) Most Confused Classes and Effect of Grayscale

Most confused classes: Looking at the confusion matrices, both models mostly confuse eosinophil and monocyte (25 errors between them).

Why grayscale makes this harder: In the original colour images, different white blood cells have very clear dye/stain colours—for example, eosinophils have bright pink/orange cytoplasm granules. When we convert the images to grayscale, all that colour difference is thrown away, leaving only cell size and lobe shapes, which look very similar at 128x128 resolution.

## (3) AI Log Table

| ***Prompt (One Line)*** | ***What I Changed*** | ***How I Checked the Result Was Correct*** |
|---|---|---|
| "Write Python code to filter BloodMNIST for labels [1, 5, 6, 7], sample 100/30/30 per class with seed, and resize to 128x128 grayscale." | Set REG_NO = "017" (SEED = 17), mapped old labels [1, 5, 6, 7] to 0..3, and used cv2.resize with IMG = 128. | Printed the X and y shapes ((400, 128, 128), (120, 128, 128)) and class counts per split, and plotted a 4x4 image grid. |
| "How to extract HOG features in scikit-image and tune LinearSVC with C in [0.1, 1, 10] on validation set" | Changed pixels_per_cell to (16, 16) to match my odd registration number rule and added StandardScaler in make_pipeline. | Checked that the printed HOG feature vector length matched my hand calculation (1764) and plotted the HOG images for 4 classes. |
| "Create a fixed 6x6 grid of OpenCV KeyPoints on a 128x128 image with size 32 and compute SIFT descriptors" | Adjusted the grid step spacing so all 36 keypoints sit evenly inside the 128x128 image and flattened the output array. | Printed the output descriptor shape per image to verify it was 36x128 = 4608 and checked that no None descriptors were returned. |
| "Write functions to rotate a 128x128 grayscale image by 20 degrees and add Gaussian noise with sigma 0.05" | Scaled the image to [0, 1] before adding noise (sigma = 0.05), clipped values back to [0, 1], and converted back to uint8 [0, 255]. | Plotted sample clean, rotated, and noisy test images side by side and checked dtype=uint8 before extracting HOG and SIFT features. |